

DEEPAK KUMAR SARANGI

Senior Software Engineer | Backend Systems & Distributed Platforms

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PROFESSIONAL SUMMARY

Senior Software Engineer with 10+ years of experience building scalable backend systems and distributed platforms in Java. At Walmart, designed and owned high-throughput backend services and event-driven architectures processing billions of events daily on GCP — cutting data delivery latency by 90%. Known for strong system design instincts, cross-team technical leadership, and applying big data tooling (Spark, Kafka, Airflow) where it serves the backend architecture. Targeting Staff / Lead Engineer roles where I can own platform design, define engineering standards, and lead high-impact backend initiatives.

EXPERIENCE

Walmart | Senior Software Engineer

Feb 2022 – Present

Backend Systems & Platform Architecture

- Designed and owned end-to-end backend architecture for distributed, event-driven systems processing 1–10 billion events/day on GCP — defined SLAs, failure handling, and scalability patterns (Pub/Sub, Dataflow, BigQuery, Bigtable) adopted across 3+ downstream teams.
- Architected a high-throughput streaming backend (Kafka + Spark) that reduced data delivery latency from 3 hours to 30 minutes (90% improvement), directly unblocking downstream ML and analytics consumers.
- Built a Spring Boot–based multi-tenant backend platform for large-scale data distribution, integrating GCS, Hive, and SFTP/API clients with resiliency patterns (circuit breakers, retries, dead-letter queues) for fault-tolerant, high-volume data transfer.
- Owned 5–10 production backend services and processing systems end-to-end; optimized Spark job runtime by 20% through query tuning, partitioning strategy, and execution plan analysis.
- Implemented data validation and quality enforcement systems, reducing data defects by 25% across pipelines.
- Built end-to-end pipeline observability using Prometheus and Grafana — heartbeat-based alerting reduced mean time to detection of DAG failures and SLA breaches across all Airflow-orchestrated workflows.
- Managed and operated Airflow workflows at scale via Astronomer, ensuring reliable orchestration of mission-critical backend processing jobs with high availability and on-call SLA adherence.

GenAI & LLM Tooling

- Built an end-to-end LLM-powered developer productivity agent that converts Jira requirements into production-ready PRs: scans requirements → plans → generates code → builds → auto-fixes CI failures → deploys → runs tests → raises a merge-ready PR with zero manual handoff.
- Designed an LLM-powered cost intelligence tool for internal teams: scans AI/LLM project codebases, detects cost-leakage patterns (context rot, suboptimal prompt compression, oversized RAG fetch windows, high-cost model overuse), quantifies potential savings, validates fixes, and raises merge-ready PRs in target codebases.
- Built multi-tenant chatbot architecture with tenant isolation and shared services, reducing infrastructure cost by 40% while improving self-service operations.

Technical Leadership & Cross-Team Influence

- Authored RFCs and system design documents for 3+ cross-team initiatives; standards adopted across CRM and data platform organizations.
- Drove cross-team technical decisions on distributed system design, API contracts, and service reliability — key voice in architecture and design reviews across squads.
- Mentored 3–5 engineers across experience levels; led hiring panels and technical interviews.
- Held end-to-end ownership over 2 of 5 core applications within the team; primary design decision partner for the team lead, contributing to architecture and technical direction across all 5 apps.
- Presented internal tech forums on streaming platform design and LLM-powered developer tooling.

Fidelity Investments | Lead Software Engineer

May 2021 – Feb 2022

- Contributed to design and development of a pricing engine supporting batch and near-real-time data processing for financial instruments.
- Integrated real-time data feeds from multiple upstream sources, improving data freshness and system responsiveness.

- Implemented observability stack with metrics, dashboards, and log analysis (Prometheus/Grafana), improving incident detection and response time.

Tata Consultancy Services (TCS) | IT Analyst

Mar 2016 – May 2021

Progression: Assistant System Engineer → System Engineer → IT Analyst | 3 client projects across multiple domains

- Designed and built Java backend services from scratch across 3 client projects spanning multiple domains, taking ownership of requirements through to deployment on each engagement.
- Developed and maintained enterprise middleware and integration layers, enabling reliable communication across heterogeneous systems in large-scale client environments.
- Identified and resolved production incidents independently through structured root cause analysis, improving system stability and reducing recurrence across integration workflows.
- Drove performance improvements in backend integration workflows, reducing processing latency and improving throughput for downstream consumers.
- Mentored and onboarded junior engineers joining the team, providing technical guidance on Java development, integration patterns, and debugging practices.
- Progressed through three internal roles over 5 years — a recognition of consistent delivery, cross-project adaptability, and growing ownership across client engagements.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, SQL

Backend & APIs: Spring Boot, RESTful APIs, Microservices, Distributed Systems, Event-Driven Architecture

Databases: PostgreSQL, MySQL, Cassandra, Cosmos DB

Big Data & Streaming: Apache Spark, Apache Kafka, Apache Airflow, Astronomer, Pub/Sub, Dataflow

Cloud (GCP): BigQuery, Bigtable, Cloud Storage, Dataflow, Pub/Sub, Dataproc, GCS

GenAI & LLM: LangChain, OpenAI APIs, Claude, Ollama, Prompt Engineering, RAG

Observability: Prometheus, Grafana, Splunk, Openobserve

DevOps & CI/CD: Docker, Kubernetes, Jenkins, Concord, Git

EDUCATION

Bachelor of Technology, Mechanical Engineering | GCEK, Bhawanipatna | CGPA: 7.7